

Joshua Clavecillas

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Physics • Applied Mathematics • California State University San Bernardino
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Education

California State University, San Bernardino 2024–2028

B.S. Physics • B.S. Applied Mathematics (Double Major)

Physics Coursework: Computational Physics, Mathematical Methods, Intermediate Laboratory, Statistical Mechanics, Electrodynamics

Chaffey College 2022–2024

Physics Coursework: General Physics I–III, Differential Equations, Calculus I–III

Research Interests

Dark Matter Instrumentation • Computational Physics • Machine Learning for Physical Systems • Experimental E&M • Condensed Matter Physics • Spectroscopy and Magnetometry

Research Experience

Detection Physics DOE Research Intern Summer 2026

Pacific Northwest National Laboratory

- Worked with an axion dark matter haloscope detector system, with emphasis on resonant-cavity tuning, detector characterization, and hardware-adjacent data workflows.
- Research direction connects upper-division E&M, microwave/RF detector behavior, numerical simulation, and data acquisition.
- Built a workflow that takes in raw data from a vector network analyser, characterizes and extracts meaningful parts of the data, to then train a NN that predicts the geometry of the cavity and its resonant frequencies off of brand new data the model has never seen before.

Computational Physics Projects

Spring 2026

CSUSB Department of Physics

- Developed a finite-difference time-domain electromagnetic radiation solver for a rotating dipole model, using explicit leapfrog/Yee-style time stepping on a 2D grid. Computed and visualized Poynting-vector flow, near-field and far-field behavior, and angular intensity patterns to connect numerical fields with physical radiation observables.
- Implemented finite-difference solvers for Laplace, Poisson, and wave-equation problems, including centered stencils, residual tracking, boundary conditions, and convergence diagnostics.
- Modeled nonlinear Lotka–Volterra dynamics with large parameter sweeps and reinforcement learning used for classification of coexistence, oscillatory, and extinction regimes.

Intermediate Laboratory Experiments

Spring 2026

CSUSB Department of Physics

- Refined a Michelson interferometer experiment for refractive-index measurements of glass, including fringe counting, angle sweeps, optical alignment, and uncertainty propagation. Improved interferometer stability and fringe visibility through optics-table setup, optics cleaning, and noise-reduction procedures.
- Performed PPMS VersaLab vibrating-sample magnetometry on magnetite (Fe_3O_4), analyzing $M(H)$ hysteresis and $M(T)$ behavior including Verwey-transition signatures near low temperature.
- Gained experience with spectroscopy, optical instrumentation, oscilloscopes, data acquisition, calibration practices, and statistical analysis including weighted fits and chi-squared checks.

Poster and Oral Presentations

July Intern Research Symposium

July 2026

Pacific Northwest National Laboratory

- Clavecillas, J., *Physics Informed Neural Network Approach to Resonant Cavity Characterization and Mode Identification in an Axion Haloscope*. Poster presented at the July Intern Research Symposium at PNNL
- Clavecillas J., *Cavity Resonator Mode Finding Algorithm*. Oral presentation to the ADMX Cavity Call.

Awards and Honors

Cal-Bridge Scholar

2026–present

Cohort 13

- Selective California State University program preparing undergraduates for Ph.D. study in physics, astronomy, and related STEM fields through mentorship, research opportunities, and professional development.

Teaching, Tutoring, and Leadership

President, Society of Physics Students (SPS)

2026–Present

California State University, San Bernardino

- Lead chapter programming, peer mentorship, research engagement, and professional-development activities for undergraduate physics students.
- Organize student events, outreach efforts, and department-facing initiatives to strengthen the CSUSB physics community.

STEM Tutor (Math & Physics)

2022–Present

- Tutored students in calculus, linear algebra, classical mechanics, electromagnetism, thermodynamics, differential equations, and introductory physics.
- Designed problem sets, conceptual explanations, and exam-prep outlines for high-school and college students, with emphasis on translating equations into physical reasoning.

STEM Club Founder and President — Chaffey College

2022–2024

- Organized peer support, STEM community-building, and academic development activities for transfer-oriented science and engineering students.

Mathnasium Instructor / Assistant Center Director

2022–2026

- Led math tutoring, student progress support, and center operations while mentoring students across algebra, geometry, precalculus, and calculus preparation.

Technical Skills

Programming and Scientific Computing: Python (NumPy, pandas, scikit-learn, PyTorch, TensorFlow), MATLAB, Mathematica, L^AT_EX, Git/GitHub.

Simulation and Modeling: COMSOL Multiphysics (end-to-end project design, CAD geometry, FEA, EM wave eigenfrequency and eigenmode analysis); finite-difference time-domain (FDTD) electromagnetic solvers; finite-difference solvers for Laplace, Poisson, and wave-equation problems with convergence diagnostics.

Machine Learning: Gradient-boosted decision trees, feedforward and convolutional neural networks, supervised regression and classification, feature engineering, hyperparameter tuning, validation, and model-performance evaluation. Applied to resonant-peak recognition, cavity geometry and resonant-frequency prediction from vector network analyzer data, elemental concentration prediction from ICP-MS measurements, and detector-response modeling.

Laboratory and Instrumentation: RF/microwave detector systems, vector network analyzers, Michelson interferometry, optics-table alignment, spectroscopy, oscilloscopes, data acquisition, PPMS VersaLab VSM magnetometry, calibration practices.

Data Analysis: Uncertainty propagation, weighted least squares, chi-squared analysis, curve fitting, covariance and correlation analysis, experimental error analysis.

CAD and Design: Mechanical part design in CAD; designed and fabricated components adopted in workflow and efficiency improvements at prior employers.

Professional Development

High Energy Physics AI Bootcamp

June 2026

Pacific Northwest National Laboratory

- Developed a gradient-boosted decision-tree regression model using scikit-learn to predict elemental concentrations from ICP-MS measurements.
- Built feedforward neural-network models using PyTorch and TensorFlow for automated recognition of resonant peaks in spectral data.
- Extended the peak-recognition workflow into a convolutional neural network to extract localized spectral features directly from graphical or array-based inputs.
- Applied data preprocessing, feature engineering, model training, validation, hyperparameter tuning, and scientific visualization to high-energy-physics and instrumentation datasets.